

HEX20 — Final Technical Demonstration Report

Onboard Telecommand Reception & Routing — Milestone 11

Scope

This report packages the secured telecommand reception, validation, routing, simulation, dashboard and final validation work completed across Milestones 1–10.

Pipeline

Noisy uplink → frame synchronization → transfer-frame decoding and CRC → Space Packet decoding → security gate → APID/service/subtype validation → PUS handler → telemetry/dashboard.

Command coverage

Command	APID	Service	Subtype	Role
HKR	100	3	1	Housekeeping
MEM	101	6	2	Memory
FUN	102	8	128	Function
CON	103	17	1	Connection

M10 validation campaign

The final stress campaign contains 1,000 deterministic scenarios covering normal commands, noisy uplink recovery, CRC faults, authentication tampering, header manipulation, unknown authenticated commands, malformed security envelopes, malformed packets and replay attempts.

Requirements traceability

ID	Requirement	Verification
HEX20-TC-001	Receive noisy uplink stream	M3/M8/M10
HEX20-TC-002	Decode transfer frame	M1/M8
HEX20-TC-003	Validate CRC	M1/M3/M8/M10
HEX20-TC-004	Extract space packet	M1/M8/M10
HEX20-TC-005	Route APIDs	M2/M7
HEX20-TC-006	Handle PUS services	M2/M8/M9
HEX20-TC-007	Security authentication	M5/M7/M8/M10
HEX20-TC-008	Replay protection	M5/M8/M10
HEX20-TC-009	Header binding	M8/M10
HEX20-TC-010	Telemetry/metrics	M4/M9
HEX20-TC-011	Prototype latency	M8/M10
HEX20-TC-012	Demonstration readiness	M11

Demonstration procedure

1. Start the M9 Streamlit dashboard.
2. Transmit HKR, MEM, FUN and CON and show accepted routing.
3. Inject authentication tampering and show SECURITY rejection.
4. Inject CRC corruption and show FRAME rejection.
5. Inject header mismatch and show ROUTING rejection.
6. Explain replay protection and show M10 validation evidence.

Limitations and next engineering steps

This is a software prototype and does not establish flight readiness, cryptographic certification, radiation tolerance, deterministic hardware timing or formal standards compliance. Production work should use mission-approved cryptography and key management, exact CCSDS/PUS profiles, persistent replay-state handling, hardware-in-the-loop testing and formal verification/qualification where required.